








Advanced Time Series Analysis and Forecasting

Seminar 3: Structural VAR and local projections



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Acronyms used in this material (1/2)

- **AAA**: highest credit rating (AAA-rated euro-area government bonds)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BQ**: Blanchard–Quah (long-run identification)
- **CPI**: Consumer Price Index
- **EA**: Euro Area
- **EBP**: Excess Bond Premium (Gilchrist–Zakrajšek)
- **ECB**: European Central Bank
- **EIA**: US Energy Information Administration
- **FEVD**: Forecast Error Variance Decomposition
- **FF4**: surprise in the federal funds futures contract three months ahead
- **FOMC**: Federal Open Market Committee

Acronyms used in this material (2/2)

- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **GNP**: Gross National Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **IP**: Industrial Production
- **IRF**: Impulse Response Function
- **LP**: Local Projection
- **LP-IV**: Local Projection with Instrumental Variables
- **MA**: Moving Average
- **MSE**: Mean Squared Error
- **OLS**: Ordinary Least Squares
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **SVAR**: Structural Vector AutoRegression
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression

Today's question and seminar structure

- The question of the seminar
  - **how do we turn the correlations of a VAR into the effects of identified shocks, and how much should we trust them?**
- Outputs to produce
  - the answers to the A exercises
  - one figure and one table for each B exercise
  - the analysis requested in C1 and the verdicts on the AI answer in C2, each with one line of justification
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the exercise map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	Cholesky by hand; a non-recursive system with a known elasticity	Solved, Proposed	A1
A3, A4	a long-run restriction; the set identified by signs	Solved, Proposed	A3
A5, A6	an external instrument; local projections in an AR(1)	Solved, Proposed	A5
A7, A8	variance decomposition; identification through heteroskedasticity	Solved, Proposed	A7
B1, B2	Kilian (2009) on oil; orderings of a Romanian VAR	Solved, Proposed	B1
B3–B5	Gertler–Karadi proxy SVAR; Romer–Romer by LP and VAR; Blanchard–Quah robustness	Solved, Proposed	B1, B3
C1, C2	euro-area spillovers with a weak instrument; what is wrong in an AI answer?	Proposed	B3, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; its solution is discussed in the seminar
- **A**: calculations and derivations on paper; **B**: real data, solved in the notebook, each closing with an interpretation question; **C**: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
oil production, real activity, real oil price	EIA, FRED	monthly	1974–2026
RO and EA IP, HICP, 3-month rates; EUR/RON	Eurostat, BNR	monthly	2005–2026
US macro data, FF4, Romer–Romer shocks	replication files of Ramey (2016)	monthly	1965–2015
real GNP, unemployment of men 20+	FRED	quarterly	1948–2026
ECB policy shocks; EA 1-year rate	Jarociński and Karadi (2020); ECB	monthly	2005–2025

- In the notebook: `var_ols`, `irf`, `fevd`, `bq_impact`, `bootstrap_var`, `proxy_irf`, `proxy_mbb`, `lp`, `lp_iv`; no account or key is needed

Prerequisites for Today (1/23): the reduced-form VAR

- Details: Chapter 3
- **VAR**(p) in reduced form (TSA, Chapter 6): each variable on p lags of all variables

$$y_t = \nu + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t, \qquad \mathbb{E}u_t = 0, \quad \mathbb{E}u_t u_t' = \Sigma_u$$

  - y_t: n variables; ν: intercepts; A_j: $n \times n$ coefficient matrices
  - u_t: the one-step forecast errors (innovations), correlated across equations, with covariance Σ_u
  - estimation: OLS equation by equation; $\hat{\Sigma}_u = T^{-1} \sum_t \hat{u}_t \hat{u}_t'$
  - stable if all eigenvalues of the companion matrix lie inside the unit circle: shocks die out

Prerequisites for Today (2/23): choosing the lag length

- Information criteria for $p = 0, 1, \ldots, p_{\max}$

$$\mathrm{AIC}(p) = \ln |\hat{\Sigma}_u(p)| + \frac{2pn^2}{T}, \qquad \mathrm{BIC}(p) = \ln |\hat{\Sigma}_u(p)| + \frac{pn^2 \ln T}{T}$$

  - $|\hat{\Sigma}_u(p)|$: the determinant of the residual covariance with p lags (it falls as p grows)
  - pn^2: the number of slope coefficients; choose the p with the smallest value
  - BIC penalises more ($\ln T > 2$ for $T \geq 8$): it chooses fewer lags than AIC
- Same sample for every p
  - start all fits at observation $p_{\max} + 1$, otherwise the criteria compare different data

Prerequisites for Today (3/23): impulse responses

- **MA representation**: the VAR as a sum of past innovations

$$y_t = \mu + \sum_{h=\mathbf{0}}^{\infty} \Phi_h u_{t-h}, \qquad \Phi_{\mathbf{0}} = I, \quad \Phi_h = \sum_{j=\mathbf{1}}^{\min(h,p)} A_j \Phi_{h-j}$$

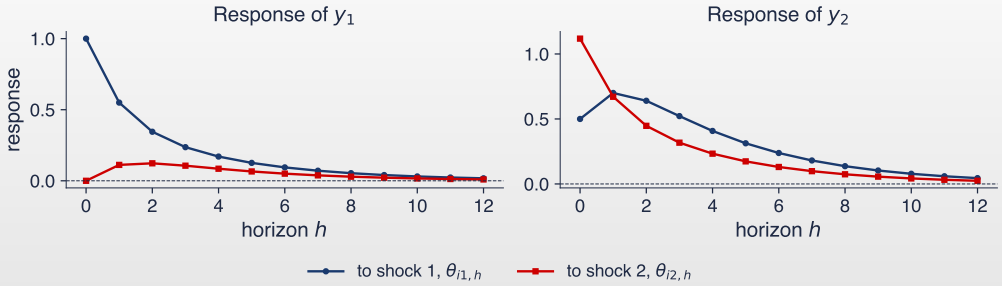
  - Φ_h: the effect of an innovation after h periods; μ: the mean of y_t

- **Structural** shocks ε_t with $u_t = B_{\mathbf{0}} \varepsilon_t$: the **impulse responses** (IRF)

$$\Theta_h = \Phi_h B_{\mathbf{0}}, \qquad \theta_{ij,h} = \frac{\partial y_{i,t+h}}{\partial \varepsilon_{jt}}$$

  - $\theta_{ij,h}$: the response of variable i to shock j after h periods; $\Theta_{\mathbf{0}} = B_{\mathbf{0}}$: the impact effects
  - cumulated responses $\sum_{s \le h} \theta_{ij,s}$: the response of the level when y_i is a growth rate

Prerequisites for Today (4/23): impulse responses of a VAR(1)



- A bivariate VAR(1) with chosen coefficients; shocks identified by Cholesky (next slides)
- Ordered second, y_2 cannot move y_1 on impact: $\theta_{12,0} = 0$; afterwards it can, through the lags
- All responses die out: the VAR is stable

Prerequisites for Today (5/23): the identification problem

- Structural shocks: uncorrelated, unit variance, $\mathbb{E}\varepsilon_t\varepsilon_t' = I$

$$u_t = B_0\varepsilon_t \quad \Longrightarrow \quad \Sigma_u = B_0 B_0'$$

  - B_0: column j holds the impact of shock j on the n variables
- **Counting**: Σ_u is symmetric
  - $n(n+1)/2$ distinct moments for n^2 unknowns in B_0: $n(n-1)/2$ restrictions are needed
  - **observational equivalence**: $B_0 Q$ with Q orthogonal ($QQ' = I$) gives the same Σ_u, hence fits the data equally well
- Sources of restrictions
  - timing (zeros on impact), a known elasticity from outside the data Blanchard and Perotti (2002), long-run zeros, signs, changes in volatility, external instruments

Prerequisites for Today (6/23): the Cholesky identification

- **Cholesky**: $B_0 = P$, the unique lower-triangular matrix with positive diagonal and $PP' = \Sigma_u$

$$P = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix}, \qquad p_{11} = \sqrt{\sigma_{11}}, \quad p_{21} = \frac{\sigma_{21}}{p_{11}}, \quad p_{22} = \sqrt{\sigma_{22} - p_{21}^2}$$

  - σ_{ij}: the elements of Σ_u
  - a **recursive** ordering: variable i reacts within the period only to shocks $1, \ldots, i$; a timing assumption, so the ordering matters

- **Scaling** a shock
  - a one-standard-deviation shock j has impact column b_j; to report a shock that raises variable k by c (e.g. 25 bp), use $c\, b_j / b_{kj}$
  - bp: basis point, 0.01 percentage points

Prerequisites for Today (7/23): the variance decomposition

- h-step forecast error of $y_{i,t+h}$: a sum of future structural shocks

$$y_{i,t+h} - \mathbb{E}_t y_{i,t+h} = \sum_{s=0}^{h-1} \sum_{k=1}^{n} \theta_{ik,s}\, \varepsilon_{k,t+h-s}$$

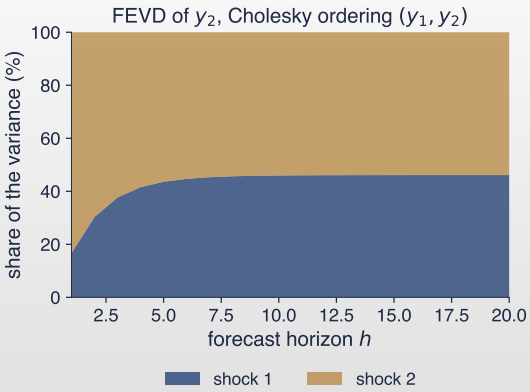
  - the shocks are uncorrelated with unit variance, so the variance is $\sum_{s<h} \sum_k \theta_{ik,s}^2$

- **FEVD** (forecast error variance decomposition): the share of shock j

$$\omega_{ij,h} = \frac{\sum_{s=0}^{h-1} \theta_{ij,s}^2}{\sum_{s=0}^{h-1} \sum_{k=1}^{n} \theta_{ik,s}^2}$$

  - the shares sum to 1 over j; $h = 1$ uses only the impact matrix B_0; as $h \to \infty$ they converge for a stable VAR

Prerequisites for Today (8/23): reading a variance decomposition



- The VAR(1) of the previous chart, variable y_2
- At $h = 1$ shock 1 already explains part of y_2: the innovations are correlated and y_1 is ordered first
- Its share grows with h, because y_1 feeds into y_2 through the lags
- A different ordering gives different shares: the FEVD inherits the identification

Prerequisites for Today (9/23): long-run restrictions

- **Blanchard–Quah** Blanchard and Quah (1989): restrict the cumulated (long-run) responses

$$\Theta(1) = \sum_{h=\mathbf{0}}^{\infty} \Theta_h = A(1)^{-\mathbf{1}} B_{\mathbf{0}}, \qquad A(1) = I - \sum_{j=\mathbf{1}}^{p} A_j$$

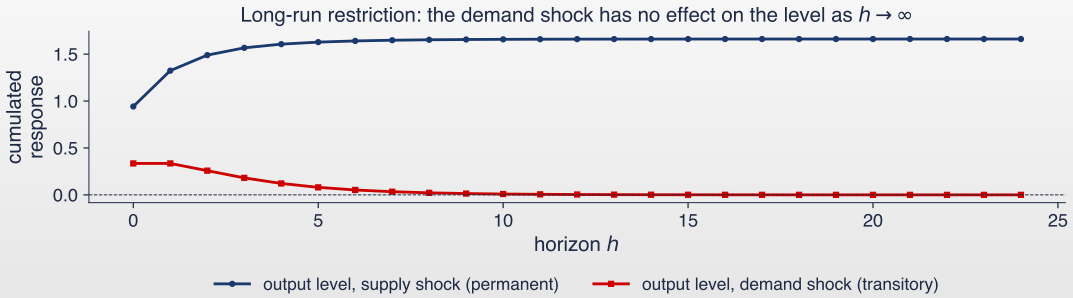
  - $\Theta(1)$: the long-run effect of each shock on the variables; when y_i is a growth rate, on the level of y_i
  - a zero in $\Theta(1)$: a shock with no permanent effect (BQ: demand does not move the level of output in the long run)

- Computation with $\Theta(1)$ lower triangular

$$\Theta(1) = \mathrm{chol}\big(A(1)^{-\mathbf{1}}\, \Sigma_u\, A(1)^{-\mathbf{1}\prime}\big), \qquad B_{\mathbf{0}} = A(1)\,\Theta(1)$$

  - chol: the lower-triangular Cholesky factor; the first equation: $\Theta(1)\Theta(1)'$ is the long-run covariance of y_t
  - $A(1)$ is estimated imprecisely when the series are persistent: the long-run answer is fragile

Prerequisites for Today (10/23): permanent and transitory shocks



- A VAR(1) in output growth and unemployment, identified by the long-run zero
- The supply shock moves the output level permanently; the demand shock moves it for a while, then the level returns to 0

Prerequisites for Today (11/23): sign restrictions

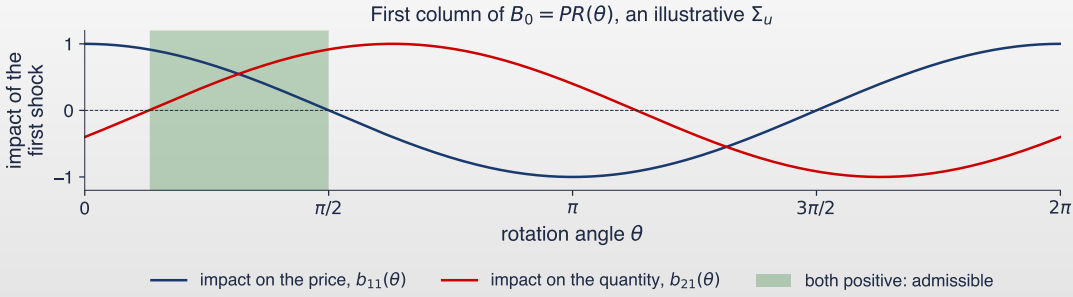
- **Sign restrictions** Uhlig (2005), Rubio-Ramírez, Waggoner and Zha (2010): every admissible B_0 is PQ, with P the Cholesky factor and Q orthogonal

$$n = 2: \quad Q = R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \theta \in [0, 2\pi)$$

  - $R(\theta)$: a rotation by the angle θ; every θ reproduces Σ_u exactly

- Algorithm
  - draw Q (in 2×2: θ on a grid or at random); compute the responses of PQ
  - keep Q if the responses have the required signs (e.g. demand raises both price and quantity); else discard
  - the result is an identified **set** of B_0 and of responses, not a single point

Prerequisites for Today (12/23): the set identified by signs



- Each angle θ gives one candidate first column of B_0; all of them fit Σ_u
- Green: the angles where both impacts are positive; the impacts vary inside this region, so they are only set-identified

Prerequisites for Today (13/23): identification through heteroskedasticity

- **Rigobon** Rigobon (2003): two volatility regimes, the same impact matrix

$$\Sigma_1 = B_0 B_0', \qquad \Sigma_2 = B_0 \Lambda B_0', \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$$

  - Σ_1, Σ_2: the innovation covariances in regimes 1 and 2; λ_i: the variance of shock i in regime 2 relative to regime 1

- Computation

$$\Sigma_1^{-1}\Sigma_2 = (B_0')^{-1}\Lambda B_0'$$

  - the λ_i are the eigenvalues of $\Sigma_1^{-1}\Sigma_2$; the columns of B_0 follow from its eigenvectors, rescaled so that $B_0 B_0' = \Sigma_1$
  - two regimes give $n(n+1)$ moments for $n^2 + n$ unknowns (B_0 and Λ)

Prerequisites for Today (14/23): external instruments

- **Proxy** (external instrument) z_t Stock and Watson (2018), Mertens and Ravn (2013): an observed variable related to shock 1 only

$$\text{relevance}: \ \mathbb{E} z_t \varepsilon_{\mathbf{1}t} = \alpha \neq 0, \qquad \text{exogeneity}: \ \mathbb{E} z_t \varepsilon_{jt} = 0, \ j \neq 1$$

  - then $\mathbb{E} u_t z_t = B_{\mathbf{0}} \mathbb{E} \varepsilon_t z_t = \alpha b_{\mathbf{1}}$: the covariances of the innovations with z_t are proportional to the first column of $B_{\mathbf{0}}$

- **Relative impacts** (the scale α cancels)

$$\frac{b_{i\mathbf{1}}}{b_{\mathbf{11}}} = \frac{\text{Cov}(u_{it}, z_t)}{\text{Cov}(u_{\mathbf{1}t}, z_t)}$$

  - normalise: a shock that raises variable 1 (e.g. the policy rate) by 25 bp has impacts $0.25\, b_{i\mathbf{1}}/b_{\mathbf{11}}$
  - only shock 1 is identified; the other columns of $B_{\mathbf{0}}$ stay unknown

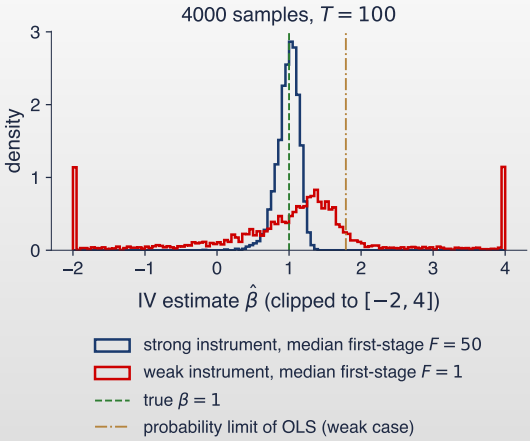
Prerequisites for Today (15/23): first stage and weak instruments

- **First stage**: regress the innovation of the policy variable on the instrument

$$u_{1t} = \pi z_t + e_t, \qquad F = \left(\frac{\hat{\pi}}{\mathrm{se}(\hat{\pi})}\right)^2$$

  - $\mathrm{se}(\hat{\pi})$: robust (heteroskedasticity- or HAC-) standard error; with one instrument F is the squared t-statistic
  - rule of thumb: $F < 10$ signals a **weak instrument**
- With a weak instrument
  - the IV estimate is biased towards OLS, its distribution is far from Normal and t-tests over-reject
  - IV: instrumental variables; with one instrument $\hat{\beta}_{IV} = \widehat{\mathrm{Cov}}(z, y)/\widehat{\mathrm{Cov}}(z, x)$
- **High-frequency surprises**: FF4 = change of the three-month-ahead federal funds futures rate in a 30-minute window around an FOMC announcement Gertler and Karadi (2015)
  - ECB analogue: the pure monetary policy shocks of Jarociński and Karadi (2020)

Prerequisites for Today (16/23): strong and weak instruments



- Simulated IV estimates of $\beta = 1$ with an endogenous regressor
- Strong instrument
  - the estimates are centred on 1 and concentrated
- Weak instrument
  - wide, skewed, pulled towards the OLS limit
  - a t-test built on them is not reliable

Prerequisites for Today (17/23): the Anderson–Rubin set

- **Anderson–Rubin** Montiel Olea, Stock and Watson (2021): test each candidate effect β_0 of x on y with instrument z

$$y_t - \beta_0 x_t = g\, z_t + e_t, \qquad \mathrm{AR}(\beta_0) = \left(\frac{\hat{g}}{\mathrm{se}(\hat{g})}\right)^2$$

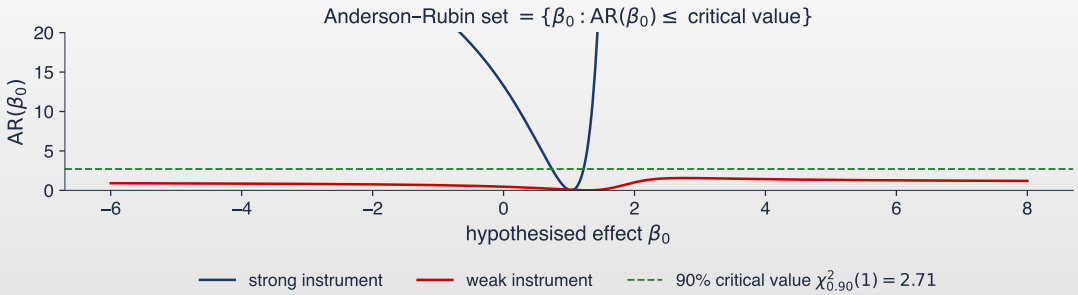
  - if β_0 is the true effect, $y_t - \beta_0 x_t$ is unrelated to z_t, so $g = 0$; $\mathrm{AR}(\beta_0) \sim \chi^2(1)$ however weak z is

- Confidence set at level $1 - \alpha$

$$\mathcal{C}_{1-\alpha} = \{\beta_0 : \mathrm{AR}(\beta_0) \leq \chi^2_{1-\alpha}(1)\}$$

  - computed on a grid of β_0; it can be an interval, two half-lines or the whole real line
  - an unbounded set says that the data do not determine the effect

Prerequisites for Today (18/23): a bounded and an unbounded AR set



- Blue (strong instrument): the statistic rises steeply on both sides; the set is a short interval
- Red (weak instrument): the statistic stays below the line over most of the grid; the set is unbounded

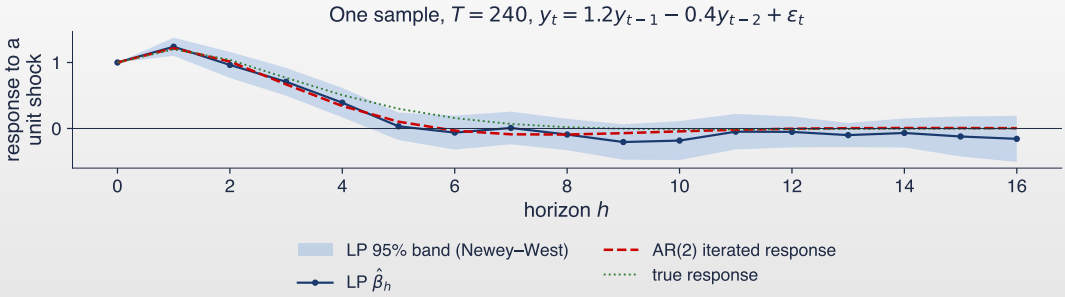
Prerequisites for Today (19/23): local projections

- **Local projections** (LP) Jordà (2005): one regression per horizon h

$$y_{t+h} = \mu_h + \beta_h x_t + \gamma_h' w_t + \xi_{t+h}, \qquad h = 0, 1, \ldots, H$$

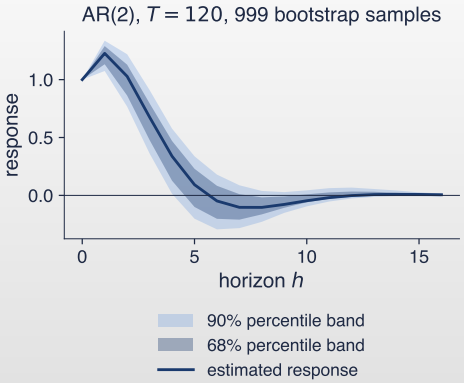
  - x_t: the shock (or the policy variable); w_t: controls (lags of y, x and other variables); ξ_{t+h}: the error
  - $\hat{\beta}_h$: the response at horizon h, read directly, without iterating a model
- Inference
  - ξ_{t+h} is autocorrelated: Newey–West standard errors with $h+1$ lags
  - or **lag augmentation**: one extra lag of the controls and heteroskedasticity-robust SE Montiel Olea and Plagborg-Møller (2021)
- **LP-IV**: x_t is the policy variable, instrumented by z_t Stock and Watson (2018)
  - with the same lags, LP and VAR estimate the same responses up to horizon p Plagborg-Møller and Wolf (2021)

Prerequisites for Today (20/23): local projections and a VAR on one sample



- A simulated AR(2) with an observed shock; LP regresses y_{t+h} on the shock and two lags of y
- The LP responses are noisier and their band widens with h; the iterated AR(2) response is smooth but relies on the model being right

Prerequisites for Today (21/23): bootstrap bands for responses



- **Recursive-design** bootstrap Gonçalves and Kilian (2004): rebuild the series from the estimated VAR

$$y_t^* = \hat{\nu} + \sum_{j=1}^{p} \hat{A}_j y_{t-j}^* + u_t^*$$

  - residual: u_t^* drawn with replacement from $\hat{u}_t$ (i.i.d. errors)
  - wild: $u_t^* = \eta_t \hat{u}_t$, $\eta_t = \pm 1$ (conditional heteroskedasticity); moving block: blocks of $\hat{u}_t$ (proxy SVAR) Jentsch and Lunsford (2019)

- Re-estimate the VAR on each y^*; **percentile band**: the quantiles of the responses at each h
  - 68%, 90% or 95%; pointwise in h, not for the whole path

Prerequisites for Today (22/23): the delta method

- **Delta method**: the standard error of a smooth function g of an estimate $\hat{\rho}$

$$g(\hat{\rho}) \approx g(\rho) + g'(\rho)(\hat{\rho} - \rho) \quad \Longrightarrow \quad \mathrm{se}\big(g(\hat{\rho})\big) \approx |g'(\rho)|\,\mathrm{se}(\hat{\rho})$$

  - g': the derivative, evaluated at the estimate in practice; a first-order Taylor expansion
  - example: an AR(1) response $g(\rho) = \rho^h$ has $\mathrm{se}(\hat{\rho}^h) \approx h|\rho|^{h-1}\mathrm{se}(\hat{\rho})$

- Vector version: $\mathrm{Var}(g(\hat{\theta})) \approx \nabla g'\, \hat{V}\, \nabla g$
  - ∇g: the gradient of g; $\hat{V}$: the covariance of $\hat{\theta}$; poor when g is very nonlinear near $\hat{\theta}$ (long horizons)

Prerequisites for Today (23/23): shocks and variables in today's tasks

- **Oil market** Kilian (2009): world oil production, global real activity, the real price of oil, ordered in this way
  - shocks: oil supply, aggregate demand, oil-specific demand; production does not react within the month to demand
- **Monetary policy**
  - **narrative** shocks Romer and Romer (2004): the part of the Fed's rate decision not explained by its own forecasts
  - **excess bond premium** Gilchrist and Zakrajšek (2012): the part of corporate credit spreads not explained by default risk
  - **price puzzle**: prices rise after a monetary tightening in a VAR, often taken as a sign that the shock is mis-identified
- Romania
  - ROBOR: the Bucharest interbank offered rate; HICP: harmonised consumer prices; IP: industrial production

A1: Cholesky by hand [Solved]

Task

- Bivariate VAR(1), $A_1 = \begin{pmatrix} 0.5 & 0 \\ 0.2 & 0.4 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}$.
- 1. Count the moments, the unknowns of B_0 and the restrictions needed.
- 2. Compute the Cholesky factor P and state the timing assumption it encodes.
- 3. Compute the responses $\Theta_1 = A_1 P$ after one period.
- Report: three counts, one matrix, one sentence, one matrix.

Solution

- 1. Moments $n(n+1)/2 = 3$, unknowns $n^2 = 4$, restrictions $n(n-1)/2 = 1$
- 2. $p_{11} = 2$, $p_{21} = 1/2 = 0.5$, $p_{22} = \sqrt{2 - 0.25}$: $P = \begin{pmatrix} 2.000 & 0.000 \\ 0.500 & 1.323 \end{pmatrix}$; variable 1 does not react to shock 2 within the period
- 3. $\Theta_1 = A_1 P = \begin{pmatrix} 1.000 & 0.000 \\ 0.600 & 0.529 \end{pmatrix}$: shock 1 still moves variable 2 after one period through a_{21}

Back Exercise list

A2: a non-recursive system with a known elasticity [Proposed]

Solution discussed in class

Task

- $u_1 = a_{12} u_2 + \sigma_1 \varepsilon_1$ (policy reacts to output), $u_2 = a_{21} u_1 + \sigma_2 \varepsilon_2$ with $a_{21} = -0.2$ known (as in Blanchard and Perotti (2002)); Σ_u of A1. Model: A1.
- 1. Show that $e_2 = u_2 - a_{21} u_1$ is a valid instrument for u_2 in the first equation.
- 2. Compute $a_{12} = \mathrm{Cov}(u_1, e_2)/\mathrm{Cov}(u_2, e_2)$, then σ_1^2 and σ_2^2.
- 3. Write B_0 and check $B_0 B_0' = \Sigma_u$; explain why this scheme is not recursive.
- Report: one argument, three numbers, one matrix.

A3: a long-run restriction by hand [Solved]

Task

- $y_t = (\Delta\text{output}_t,\, u_t)'$, VAR(1) with $A_1 = \begin{pmatrix} 0.2 & -0.3 \\ 0.1 & 0.6 \end{pmatrix}$, $\Sigma_u = \begin{pmatrix} 1 & -0.2 \\ -0.2 & 0.5 \end{pmatrix}$.
- 1. Compute $A(1)^{-1} = (I - A_1)^{-1}$ and $A(1)^{-1}\Sigma_u A(1)^{-1\prime}$.
- 2. Impose that shock 2 (demand) has no long-run effect on output and compute $\Theta(1)$ and B_0.
- 3. Read the sign of the impact of the supply shock on unemployment.
- Report: two matrices, two matrices, one sentence.

Solution

- 1. $\det(I - A_1) = 0.8 \cdot 0.4 + 0.03 = 0.35$;
  $A(1)^{-1} = \begin{pmatrix} 1.143 & -0.857 \\ 0.286 & 2.286 \end{pmatrix}$; product $= \begin{pmatrix} 2.065 & -1.127 \\ -1.127 & 2.433 \end{pmatrix}$
- 2. $\Theta(1) = \mathrm{chol}(\cdot) = \begin{pmatrix} 1.437 & 0.000 \\ -0.784 & 1.348 \end{pmatrix}$;
  $B_0 = A(1)\Theta(1) = \begin{pmatrix} 0.915 & 0.405 \\ -0.457 & 0.539 \end{pmatrix}$
- 3. The supply shock lowers unemployment on impact here; in the US data of Blanchard and Quah (1989) it raises it at first

Back Exercise list

A4: the set identified by sign restrictions [Proposed]

Solution discussed in class

Task

- Price and quantity innovations, $\Sigma_u = \begin{pmatrix} 1 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}$; $B_0 = PR(\theta)$ (Prerequisites (11/23)), first column $P(\cos\theta, \sin\theta)'$. Demand: price $+$, quantity $+$; supply: opposite signs. Model: A3.
- 1. Compute P and write the two impact responses to the demand shock as functions of θ.
- 2. Find the interval of θ for which all four signs hold.
- 3. Give the identified set of the impact effect of demand on quantity, and explain why a uniform prior on θ is not neutral.
- Report: one matrix, one interval, one interval, one sentence.

A5: an external instrument by hand [Solved]

Task

- Residuals of (policy rate, output, prices); $\mathrm{Cov}(u_t, z_t) = (0.020, -0.006, 0.010)'$; first stage of u_{1t} on z_t: $\hat{\pi} = 0.30$, robust SE 0.08.
- 1. Compute the relative impacts b_{i1}/b_{11} and the impacts of a shock that raises the rate by 25 bp.
- 2. Compute the first-stage t and F and judge relevance.
- 3. Name one way the exogeneity condition could fail for an FOMC surprise.
- Report: four numbers, two statistics, one sentence.

Solution

- 1. $b_{21}/b_{11} = -0.006/0.020 = -0.30$, $b_{31}/b_{11} = 0.50$; 25 bp: output -0.075, prices 0.125
- 2. $t = 0.30/0.08 = 3.75$, $F = t^2 = 14.1 > 10$: relevant by the rule of thumb
- 3. The information effect Nakamura and Steinsson (2018): the surprise also reveals the Fed's view of the economy, so $\mathbb{E} z_t \varepsilon_{jt} \neq 0$ for a demand shock j; prices rising after a tightening (here $+$) is a warning sign

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A6: local projections in an AR(1) [Proposed]

Solution discussed in class

Task

- $y_t = \rho y_{t-\mathbf{1}} + \varepsilon_t$, $\rho = 0.9$, $\mathrm{Var}\varepsilon_t = 1$, $T = 200$; the response at horizon $h = 4$. Model: A5.
- 1. Show that the LP of $y_{t+\mathbf{4}}$ on y_t has population coefficient $\rho^{\mathbf{4}}$ and error $\sum_{j=\mathbf{0}}^{\mathbf{3}} \rho^j \varepsilon_{t+\mathbf{4}-j}$.
- 2. Give the MA order and the variance of the LP error, and say why HAC errors are needed.
- 3. With $\mathrm{se}(\hat{\rho}) = \sqrt{(1-\rho^{\mathbf{2}})/T}$, give the delta-method SE of the plug-in $\hat{\rho}^{\mathbf{4}}$ and explain why LP is noisier.
- Report: one derivation, two numbers, two numbers, one sentence.

A7: variance decomposition by hand [Solved]

Task

- The VAR(1) and the Cholesky factor P of A1; $\Theta_0 = P$, $\Theta_1 = A_1 P$, $\Theta_2 = A_1^2 P$.
- 1. Compute the share of shock 1 in the one-step forecast error variance of variable 2.
- 2. Repeat for two and three steps.
- 3. Explain why the share of shock 2 in variable 1 stays zero at every horizon.
- Report: three shares, one sentence.

Solution

- 1. $h = 1$: $0.5^2/(0.5^2 + 1.75) = 12.5\%$
- 2. $\Theta_1 = \begin{pmatrix} 1.000 & 0.000 \\ 0.600 & 0.529 \end{pmatrix}$; $h = 2$: $(0.25 + 0.36)/(0.25 + 1.75 + 0.36 + 0.28) = 23.1\%$; $h = 3$: 27.9%
- 3. $p_{12} = 0$ and $a_{12} = 0$: variable 1 never responds to shock 2, so its forecast error contains only shock 1

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A8: identification through heteroskedasticity [Proposed]

Solution discussed in class

Task

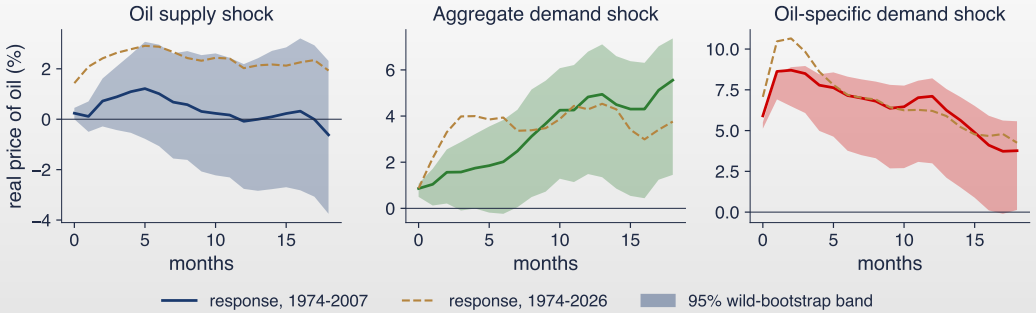
- Two regimes: $\Sigma_1 = \begin{pmatrix} 1.25 & 0.80 \\ 0.80 & 1.09 \end{pmatrix}$, $\Sigma_2 = \begin{pmatrix} 4.25 & 1.70 \\ 1.70 & 1.36 \end{pmatrix}$; B_0 is the same, the variance of shock 1 changes. Model: A7.
- 1. Show that $\Sigma_1^{-1}\Sigma_2 = B_0^{-1\prime}\Lambda B_0'$ and compute its eigenvalues.
- 2. Recover B_0 (normalise $b_{11} = b_{22} = 1$) from $\Sigma_2 - \Sigma_1$.
- 3. State the two assumptions that make this work and the one that fails if both variances change by the same factor.
- Report: two eigenvalues, one matrix, three sentences.

B1: oil supply and demand shocks [Solved]

- **Question**: does the real price of oil respond differently to supply, aggregate demand and oil-specific demand shocks, as found by Kilian (2009)?
- **Inputs**: percent change of world crude production (EIA), real activity (FRED IGREA), log real refiner acquisition cost (EIA, deflated by CPI); monthly 1974–2007 and 1974–2026; VAR(24), Cholesky in this order
- **Tasks**
  1. Estimate the VAR(24) on 1974–2007 and identify the three shocks, signing the supply shock so that it raises the price.
  2. Report the impact and 12-month responses of the real price with 95% recursive-design wild-bootstrap bands.
  3. Compute the 60-month FEVD of the price on both samples.
  4. Interpretation: which shock moves the oil price, and did the answer change after 2007?
- **Report**: six responses with bands, six variance shares, one sentence
- Notebook: section B1

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B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

Shock	impact	12 months	95% band	FEVD 60, 2007	FEVD 60, 2026
oil supply	0.24%	−0.1%	[−2.8, 2.2]	1%	9%
aggregate demand	0.86%	4.8%	[1.5, 6.8]	64%	31%
oil-specific demand	5.90%	7.1%	[3.0, 8.2]	35%	60%

- Interpretation: demand moves the price (aggregate demand with a delay, oil-specific demand at once), supply hardly at all before 2007; on the longer sample supply shocks matter more (12-month effect 2.0%) and oil-specific demand explains most of the variance

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B2: orderings of a Romanian VAR [Proposed]

- **Question**: is the rise of Romanian prices after a ROBOR shock (the price puzzle) an artefact of the ordering or of the lag length?
- **Inputs**: monthly 2005–2026: EA IP, EA HICP, Euribor 3M, RO IP, RO HICP, ROBOR 3M, EUR/RON (Eurostat, BNR), logs $\times 100$, monthly dummies; model: B1
- **Tasks**
  1. Select the lag length by BIC and AIC.
  2. For each lag length, estimate the VAR with ROBOR after RO prices, ROBOR first in the RO block and ROBOR last, keeping the euro area first.
  3. Report the 12- and 24-month responses of RO HICP, and the 12-month responses of EUR/RON and RO IP, to a 25 bp ROBOR shock.
  4. Interpretation: what does the stability of the puzzle across orderings tell you about the identification?
- **Report**: two lag lengths, a table of six rows, one sentence
- Notebook: section B2

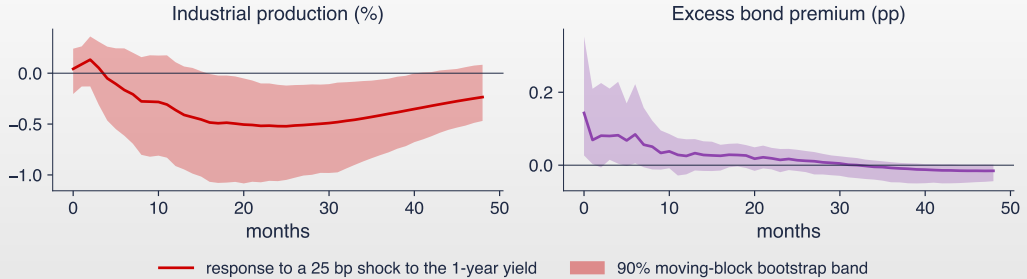
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B3: replicating Gertler and Karadi (2015) [Solved]

- **Question**: does a monetary policy shock identified with FF4 surprises raise credit spreads and lower output?
- **Inputs**: 1-year Treasury yield, log IP, log CPI, excess bond premium (Gilchrist and Zakrajšek (2012)); VAR(12), 1979:7–2012:6; instrument FF4, 1991:1–2012:6 (replication files of Ramey (2016))
- **Tasks**
  1. Estimate the reduced-form VAR and regress the yield residual on FF4 over the overlap: report the robust F.
  2. Compute the impact column from $\mathrm{Cov}(u_t, z_t)$, scaled to a 25 bp rise in the yield, and the responses for 48 months.
  3. Build 90% moving-block bootstrap bands for IP and the excess bond premium.
  4. Interpretation: through which channel does the shock work, according to these responses?
- **Report**: one F, four impacts, the IP trough with its band, one sentence
- Notebook: section B3

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B3: solution [Solved]



- First stage on 258 months: $\hat{\pi} = 1.16$, robust $F = 17.7$ (homoskedastic 21.9); impact: yield 0.250, IP 0.042%, CPI -0.042%, EBP 0.143 pp (band [0.03, 0.35])
- IP reaches -0.52% after 25 months (band $[-1.05, -0.12]$)
- Interpretation: the credit channel: the bond premium jumps at once, then output falls with a lag; the instrument is relevant ($F > 10$), in line with Gertler and Karadi (2015)

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B4: Romer–Romer shocks by LP and by VAR [Proposed]

- **Question**: do local projections and a VAR with the same narrative shock give the same output and price responses?
- **Inputs**: Romer–Romer shocks updated by Wieland (data of Ramey (2016)), log IP, unemployment, log CPI, log commodity prices, funds rate; monthly 1969:3–2007:12 ($T = 466$); model: B3
- **Tasks**
  1. Estimate LP of IP, CPI and the funds rate on the shock for $h = 0, \ldots, 48$, with 2 lags of all variables and of the shock, Newey–West SE with $h + 1$ lags.
  2. Estimate a VAR(12) with the shock ordered first and scale its responses to a 1 pp shock.
  3. Compare the trough of IP, the CPI response after 48 months and the impact on the funds rate.
  4. Interpretation: which estimator would you report, and why?
- **Report**: three pairs of numbers, two SE, one sentence
- Notebook: section B4

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B5: the robustness of Blanchard–Quah [Proposed]

- **Question**: does the share of demand shocks in output fluctuations survive changes in the treatment of low frequencies?
- **Inputs**: real GNP growth (FRED GNPC96), unemployment of men 20+ (LNS14000025), quarterly, VAR(8); model: B1
- **Tasks**
  1. Estimate the BQ model with the original treatment (break in mean growth at 1973:4, linear trend in unemployment), 1948–1987.
  2. Re-estimate with no break and no trend, and with the original treatment on data up to 2019.
  3. Report the share of demand shocks in the output-level FEVD at 4 and 40 quarters, and the long-run effect of a supply shock.
  4. Interpretation: what does the comparison say about long-run restrictions?
- **Report**: a table of three rows and four columns, one sentence
- Notebook: section B5

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C1: euro-area spillovers with a weak instrument [Proposed]

- **Question**: how much does Romanian HICP move 12 months after a 1 pp rise of the euro-area 1-year rate caused by an ECB policy shock?
- **Inputs**: Romanian data of B2, euro-area 1-year AAA spot rate (ECB), pure policy shocks of Jarociński and Karadi (2020); origins 2005:8–2025:10; model: B3
- **Tasks**
  1. Write a five-line pre-registration: outcome, horizon, instrument, controls, inference.
  2. Estimate the LP-IV at $h = 12$ with 3 lags of all variables and of the shock and monthly dummies; report $\hat{\beta}$, its Newey–West SE and the first-stage F.
  3. Build the 90% Anderson–Rubin set on a grid of $\beta \in [-15, 15]$.
  4. Interpretation: what can and cannot be concluded about ECB spillovers to Romanian prices?
- **Report**: a five-line plan, one estimate with SE, one F, one set, a project plan
- Notebook: section C1

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C2: audit an AI answer [Proposed]

- A student asked an AI assistant to summarise structural VAR identification. The answer:
- `(a) A four-variable SVAR needs four restrictions to be exactly identified.`
- `(b) In a Cholesky ordering, reordering the variables placed after the policy variable does not change the responses to the policy shock.`
- `(c) With enough horizons restricted, sign restrictions point-identify the output response.`
- `(d) In a proxy SVAR, b21/b11 equals Cov(u2, z)/Cov(u1, z).`
- `(e) Local projections are unbiased and have smaller variance than VARs, so they always dominate.`
- `(f) The wild bootstrap of Mertens and Ravn gives valid bands for proxy SVARs.`
- `(g) With Sigma = [[4, 1], [1, 2]], the Cholesky impact of shock 1 on variable 2 is 1.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- Every structural answer rests on an identifying assumption: count it, state it, defend it
- Cholesky, long-run zeros and known elasticities give points; signs give sets
- An external instrument identifies one shock if it is relevant and exogenous; check F and use robust sets
- LP and VAR answer the same question with different bias and variance
- An AI answer is a draft: check the counts, the sets, the bootstrap and the arithmetic

After the seminar

- Lecture 3 develops each topic of today: identification schemes, set identification, external instruments, inference, local projections, Romania
- C1 can grow into a team project: ECB spillovers to Romania with high-frequency instruments
- Reading: Kilian (2009); Gertler and Karadi (2015); Stock and Watson (2018); Plagborg-Møller and Wolf (2021)
- **The seminar is for practice and is not graded**

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